To appreciate the story of how physicists came to recognize the neutron-star/black-hole connection, it will help to know something about the connection in advance. Thus, the following digression:

What are the fates of stars when they die? Chapter 4 revealed a partial answer, an answer embodied in the right-hand portion of Figure 5.3 (which is the same as Figure 4.4). That answer depends on whether the star is less massive or more massive than 1.4 Suns (Chandrasekhar's *limiting mass).*

If the star is less massive than the Chandrasekhar limit, for example if the star is the Sun itself, then at the end of its life it follows the path labeled "death of Sun" in Figure 5.3. As it radiates light into space, it gradually cools, losing its thermal (heat-induced) pressure. With its pressure reduced, it no longer can withstand the inward pull of its own gravity; its gravity forces it to shrink. As it shrinks, it moves leftward in Figure 5.3 toward smaller circumferences, while staying always at the same height in the figure because its mass is unchanging. (Notice that the figure plots mass up and circumference to the right.) And as it shrinks, the star squeezes the electrons in its interior into smaller and smaller cells, until finally the electrons protest with such strong degeneracy pressure that the star can shrink no more. The degeneracy pressure counteracts the inward pull of the star's gravity, forcing the star to settle down into a white-dwarf grave on the boundary curve (white-dwarf curve) between the white region of Figure 5.3 and the shaded region. If the star were to shrink even more (that is, move leftward from the white-dwarf curve into the shaded region), its electron degeneracy pressure would grow stronger and make the star expand back to the white-dwarf curve. If the star were to expand into the white region, its electron degeneracy pressure would weaken, permitting gravity to shrink it back to the white-dwarf curve. Thus, the star has no choice but to remain forever on the white-dwarf curve, where gravity and pressure balance perfectly, gradually cooling and turning into a black dwarf—a cold, dark, solid body about the size of the Earth but with the mass of the Sun.

If the star is more massive than Chandrasekhar's 1.4-solar-mass limit, for example if it is the star Sirius, then at the end of its life it will follow the path labeled "death of Sirius." As it emits radiation and cools and shrinks, moving leftward on this path to a smaller and smaller circumference, its electrons get squeezed into smaller and smaller cells; they protest with a rising degeneracy pressure, but they protest in vain. Because of its large mass, the star's gravity is strong enough to squelch all electron protest. The